Joshua Terranova

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Curriculum Vitae

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EDUCATION

B.Sc. in Computer Science
University of Southern California, 2027

A.Sc. in Mathematics
Los Angeles Mission College, 2025

PROFESSIONAL EXPERIENCE

Lead Systems Engineer / Research Lead, NASA Team 19: *Ad Astra* *2026*
National Aeronautics and Space Administration

- Sole Lead Systems Engineer on 18-member MCA team (PM Alejandro Gonzalez, Chief Scientist Hannah Kim, DPMR Alfredo Guerra); led Engineering subteam integration with Science and Programmatic leads across MCR–PDR.
- Authored spacecraft overview, subsystem summary table (mass/power/dimensions), and top-level subsystem requirements flow-down for a lunar PSR volatile-prospecting rover.
- Owned interface architecture: 19-pathway ICD matrix, N^2 diagram, and interface narrative linking payload instruments to vehicle subsystems.
- Co-led thermal subsystem (requirements, heat-flow maps, trade study) for 40–394 K PSR environments; supported CDH trade study and communications downlink analysis.
- Owned CDH subsystem through PDR: requirements, software architecture flowchart, TRL/sub-assembly baselines, recovery/redundancy plans, and verification matrix for science data acquisition and S-band downlink.
- Researched alternative mission concepts with science lead; developed science-to-engineering traceability from volatile-detection objectives through verification planning.

Lead Systems Engineer / Research Lead, NASA Team 10: *Forged in Orbit* *2025*
National Aeronautics and Space Administration

- Sole Lead Systems Engineer on 12-member NPWEE team (PI Joel Bhattarai, PM Alexis Gallardo, Chief Engineer Justin Schueler, Chief Scientist Riya Jain); integrated Engineering, Science, and Programmatic inputs for a $10,000-class technology proposal.
- Researched thermodynamic, mechanical, and materials behavior of vacuum-environment cold welding; derived quantitative KPPs from ASTROBEAT ISS heritage, NTRS literature, and materials compatibility analysis.
- Authored cold-welding performance specifications in team Technical Research Memorandum; co-authored final proposal and quad-chart concept portfolio (cold-weld reclamation, vacuum dust ejection, triboelectric transport).
- Lead-authored L-MAP-FI: multi-agent lunar pathfinding and environmental forecasting framework (ROS2/Gazebo simulation, Fusion 360 FEA validation).
- Designed vacuum-test research protocol for weld-parameter characterization; conducted formal peer review of competing NASA proposals (Teams 16–18) on Review Panel #6.
- Lead Systems Engineer Intern, National Aeronautics and Space Administration (NASA), 2025–2026.
- Computer Engineer Intern, National Aeronautics and Space Administration (NASA), 2025 (multiple terms).

ADDITIONAL EXPERIENCE

Software Engineer, USC Field Robotics Lab *2026–Present*

University of Southern California

– Autonomy research software for University Rover Challenge 2026 on ROS 2 (Humble).
– 7-target mission orchestrator; GNSS waypoint navigation; ArUco visual servoing; YOLO detection and gating.
– Nav2 integration, EKF sensor fusion, e-stop interlocks, and judge telemetry web dashboard.

Software Engineer, USC Rocket Propulsion Lab *2025–Present*
University of Southern California

– Software systems for experimental propulsion research: design analysis, test operations, and instrumentation.

Software Engineer, USC Racing (Formula SAE) *2025–2026*
University of Southern California

Research Fellow, USC MESA *2022–Present*
University of Southern California

– Research mentorship and STEM pathway development for undergraduate research readiness and scientific communication.

CERTIFICATIONS

– NASA L'SPACE Mission Concept Academy (MCA), Certificate of Completion, Team 19: *Ad Astra*, 2026.
– NASA L'SPACE NASA Proposal Writing and Evaluation Experience (NPWEE) Academy, Certificate of Completion, Team 10: *Forged in Orbit*, 2025.

PERSONAL PROJECTS

SpaAIder (Local-First Agentic Runtime) *2025–Present*

– Foundational split: TypeScript control plane (`runOrchestrator`) owns user-facing planning and delegation; Python FastAPI substrate owns persistence, memory APIs, and service routes; Ollama supplies sovereign on-device inference across orchestrator, subagent, and embedding models.
– Memory substrate: `MemoryManager` delegates to contracted SQL tables (sessions, messages, knowledge, context, embeddings on SQLite/PostgreSQL) and an `EpisodicStore`; `/api/v1/memory/recall` fuses vector, episodic, and keyword channels before each plan.
– Episodic consolidation: session close triggers LLM synthesis of transcripts into episodes (summary, key facts, topics, tone) merged into durable user profiles.
– Context assembly: Python `ContextEngine` allocates a fixed token budget across system instructions, user profile, episodic hits, and working memory; an `EmbeddingBridge` (TF-IDF locally, Ollama `nomic-embed-text` when configured) scores similarity for recall and knowledge search.
– Ingress and observability: heterogeneous clients (web, mobile, Electron, CLI, Code-OSS IDE) POST to `/api/agent/stream`; an `AsyncEventQueue` multiplexes orchestration events—context fetch, plan emission, per-task status, subagent streams, final answer—over server-sent events.
– Planning contract: meta-orchestrator LLM receives recalled `UserContext` and must emit JSON `TaskPlan` only (intent, agent assignments, `dependsOn` DAG edges over a 135-agent type taxonomy) without executing tasks—pure delegation.
– Execution engine: dependency-aware parallel scheduler launches runnable tasks up to `maxParallelTasks`; domain subagents (code, filesystem, search, vision, memory, research, backend) run as async generators; `memory-agent` proxies read/write/recall to FastAPI; `AgentBus` provides in-process pub/sub and request–reply between agents.
– Quality loop and surfaces: critique and synthesis agents post-process subagent artifacts into one response;

completed turns persist through the memory layer; IDE fork reuses the same agent stream with editor-grounded file, selection, and terminal context.

Pneumonia Detection from Chest X-Rays *2025–2026*

- TensorFlow pipeline (`train.py`) for binary chest radiograph classification on 988 images (620 train / 368 validation).
- EfficientNetV2B0 transfer learning with focal loss, class balancing, oversampling, and two-phase fine-tuning.
- Test-time augmentation, threshold tuning, and evaluation via confusion matrices, ROC/AUC, and precision–recall curves.
- Achieved 83% validation accuracy; exported models and metrics to structured `outputs/`.

Small-Scale Language Model (LLM-V1.0) *2025–2026*

- Designed and trained a ~30M-parameter GPT-style transformer (8 layers, 8 heads, 512-dim embeddings, 8,192-token BPE vocabulary).
- PyTorch training, checkpointing, and text-generation pipelines for Colab and local workflows.

RESEARCH PROFILE

Research-focused engineer investigating lunar volatile characterization, vacuum-materials behavior, AI-driven autonomous exploration, medical image classification, and agentic AI systems. Developed SpaAIder, an independent retrieval-augmented, plan-and-execute multi-agent runtime—heterogeneous clients stream intents over server-sent events to a hierarchical orchestrator performing context hydration, meta-LLM task decomposition with dependency-aware parallel subagent scheduling, critique–synthesis refinement, Ollama-first sovereign inference, tiered episodic–semantic–vector memory, and a Code-OSS IDE fork with editor-grounded context. At NASA Team 19 (*Ad Astra*), conducted trade-study-driven research on PSR volatile prospecting at Nobile Rim 2—including thermal modeling across 40–394 K environments, science-to-requirements traceability, and subsystem trade analyses supporting in-situ volatile detection via LiDAR, spectrometry, and rotary-percussive sampling. At NASA Team 10, authored literature-backed technical research on vacuum cold-welding parameters (ASTROBEAT ISS heritage, NTRS sources, materials compatibility) and lead-authored L-MAP-FI, a multi-agent pathfinding and environmental forecasting research concept validated through ROS2/Gazebo simulation and FEA. Independently developed a chest X-ray pneumonia detection pipeline using EfficientNetV2B0 transfer learning with focal loss, class-balanced training, and test-time augmentation, achieving 83% validation accuracy on 368 held-out radiographs. Research record spans agent orchestration, persistent memory systems, autonomous rover software, deep-learning experimentation, and formal peer review of competing NASA technology proposals.

RESEARCH INTERESTS

Artificial intelligence and machine learning; inference models; retrieval-augmented generation; computer vision; medical image analysis and diagnostic machine learning; agentic AI and multi-agent orchestration; local LLM inference and transformer architectures; agentic AI with surrogate specializing models; optimization for accuracy and efficiency on orchestrator (router) LLMs in determining the single most accurate agent; autonomous mobile robotics and ROS 2 systems; spacecraft systems architecting; multi-body planetary surface mobility; autonomous roving platforms; subsystem-level interdependency modeling; fault-tolerant command and data handling for extra-terrestrial environments.

RESEARCH PUBLICATIONS AND TECHNICAL REPORTS

National Aeronautics and Space Administration (NASA), Team 19: Ad Astra, 2026

18-member L'SPACE MCA team. Lead Systems Engineer / Research Lead (sole LSE). Reports to PM Alejandro Gonzalez; integrates with Chief Scientist Hannah Kim and DPMR Alfredo Guerra.

1. Gonzalez, A., Guerra, A., Chang, A., Dixon, D., Zhou, E., Senteza, J., Mendez, J., Kumar, K., Richmond, R., **Terranova, J.**, Kim, H., French, S., Noor, M., Saad, S., Durkin, J., Raihan, F., Gutierrez, N., & Saxena, H., Preliminary engineering design, verification matrix, and integrated vehicle baseline for an autonomous lunar volatile-prospecting rover at Nobile Rim 2 (Preliminary Design Review, Team 19), National Aeronautics and Space Administration (NASA), 2026.

– Lead Systems Engineer contribution: owned CDH subsystem through PDR—requirements narrative, software architecture flowchart, TRL/sub-assembly baselines, recovery/redundancy plans, and verification matrix for science data acquisition, processing, and S-band downlink; coordinated Engineering subteam inputs under PM review.

2. Gonzalez, A., Guerra, A., Chang, A., Dixon, D., Zhou, E., Senteza, J., Mendez, J., Kumar, K., Richmond, R., **Terranova, J.**, Kim, H., French, S., Noor, M., Saad, S., Durkin, J., Raihan, F., Gutierrez, N., & Saxena, H., Definitive mission architecture, operational paradigm, and programmatic baseline for in-situ lunar volatile characterization (Mission Definition Review, Team 19), National Aeronautics and Space Administration (NASA), 2026.

– Lead Systems Engineer contribution: researched mission architecture and operational paradigms for in-situ volatile characterization at Nobile Rim 2; linked Artemis-class PSR science objectives to traverse, communications, and programmatic constraints with Science and Programmatic subteams.

3. Gonzalez, A., Guerra, A., Chang, A., Dixon, D., Zhou, E., Senteza, J., Mendez, J., Kumar, K., Richmond, R., **Terranova, J.**, Kim, H., French, S., Noor, M., Saad, S., Durkin, J., Raihan, F., Gutierrez, N., & Saxena, H., System requirements baseline, subsystem allocation framework, and interface control architecture for a lunar PSR exploration rover (System Requirements Review, Team 19), National Aeronautics and Space Administration (NASA), 2026.

– Lead Systems Engineer contribution: authored spacecraft overview, subsystem summary table (mass/power/dimensions), 19-interface ICD matrix, N^2 diagram, and interface narrative; co-led thermal heat-flow analysis and CDH/thermal trade studies for 40–394 K PSR operating environments.

4. Gonzalez, A., Guerra, A., Chang, A., Dixon, D., Zhou, E., Senteza, J., Mendez, J., Kumar, K., Richmond, R., **Terranova, J.**, Kim, H., French, S., Noor, M., Saad, S., Durkin, J., Raihan, F., Gutierrez, N., & Saxena, H., Conceptual elaboration of an autonomous lunar volatile-reconnaissance architecture for permanently shadowed region prospecting (Mission Concept Review, Team 19), National Aeronautics and Space Administration (NASA), 2026.

– Lead Systems Engineer contribution: co-authored alternative mission concept comparison with Chief Scientist Hannah Kim; developed conceptual architecture integrating LiDAR navigation, rotary-percussive regolith sampling, and onboard spectrometric volatile analysis for PSR reconnaissance.

National Aeronautics and Space Administration (NASA), Team 10, 2025

12-member L'SPACE NPWEE team. Lead Systems Engineer / Research Lead (sole LSE). PI Joel Bhattarai; PM Alexis Gallardo; Chief Engineer Justin Schueler; Chief Scientist Riya Jain.

5. Bhattarai, J., Schueler, J., **Terranova, J.**, Rojas, J., Gallardo, A., Ek, S., Jain, R., Concepcion, W., Jonson, M., Li, Q., Vasquez, G., & Atwell, J., Solid-state in-situ metallurgical reconstitution via autonomous cold-welding attachment for lunar infrastructure salvage (Final Proposal, Team 10: Forged in Orbit), National Aeronautics and Space Administration (NASA), 2025.

– Lead Systems Engineer contribution: co-authored $10,000-class final proposal; synthesized literature-backed cold-welding KPPs (100 MPa pressure, 10 kN force, ≥80 MPa shear strength) and autonomous in-situ repair architecture aligned to NASA Technology Taxonomy (TX4.3.3, TX4.6.1, TX12.4.1, TX12.4.6).

6. Bhattarai, J., Schueler, J., **Terranova, J.**, Rojas, J., Gallardo, A., Ek, S., Jain, R., Concepcion, W., Jonson, M., Li, Q., & Vasquez, G., Tripartite technology concept portfolio for autonomous lunar surface

operations and in-situ resource utilization (Quad Charts, Team 10), National Aeronautics and Space Administration (NASA), 2025.

– Lead Systems Engineer contribution: researched tripartite lunar ISRU and mobility concept portfolio—oxide-scrubbing cold-weld reclamation, ultrasonic vacuum dust ejection, triboelectric autonomous transport—with comparative cost and performance analysis (94% cost reduction vs. TIG welding; 1.5 g/min vacuum dust removal).

7. **Terranova, J.**, & Vasquez, G., L-MAP-FI: A multi-agent pathfinding and predictive environmental forecasting interface for autonomous lunar reconnaissance swarms (Technical Presentation, Team 10), National Aeronautics and Space Administration (NASA), 2025.

– Lead author: designed L-MAP-FI multi-agent pathfinding and predictive environmental forecasting framework for autonomous lunar reconnaissance swarms; validated through ROS2/Gazebo simulation and Fusion 360 FEA (70× coverage, 160× cost efficiency vs. state-of-the-art).

8. Schueler, J., **Terranova, J.**, Bhattarai, J., & Atwell, J., Thermodynamic, mechanical, and materials characterization of vacuum-environment cold-welding parameters for robotic lunar repair systems (Technical Research Memorandum, Team 10), National Aeronautics and Space Administration (NASA), 2025.

– Primary research author: derived cold-welding performance specifications from ASTROBEAT ISS in-orbit repair experiments, NTRS heritage data, and austenitic stainless/aluminum/copper alloy compatibility analysis across a −183 °C to +137 °C thermal envelope.

9. Bhattarai, J., Gallardo, A., Ek, S., Schueler, J., **Terranova, J.**, Rojas, J., Jain, R., Concepcion, W., Jonson, M., Li, Q., & Vasquez, G., Interdisciplinary workforce development and competency matrix for NASA-aligned technology infusion proposals (Teaming & Workforce Development Section, Team 10), National Aeronautics and Space Administration (NASA), 2025.

10. **Terranova, J.**, Formal peer evaluation of competing NASA technology proposals via NASA-structured scoring methodology (Review Panel #6, 2025). Evaluated proposals from Teams 16, 17, and 18.

– Lead Systems Engineer: conducted formal peer evaluation of competing NASA technology proposals (Teams 16–18) using NASA-structured scoring methodology on Review Panel #6.

Personal Projects

11. **Terranova, J.** (2025–2026). SpaAIder: A local-first multi-agent runtime with persistent cross-session memory and heterogeneous clients (Personal project).

12. **Terranova, J.** (2026). EfficientNetV2-based chest X-ray pneumonia classification with focal loss, class balancing, and test-time augmentation (Personal project).

13. **Terranova, J.** (2026). LLM-V1.0: Small-scale transformer language model for local and Colab training (Personal project).

14. **Terranova, J.** (2026). URC CV: University Rover Challenge autonomy software (USC Field Robotics Lab research software).